Urban Heating Backbone Line (U-HBL)

A City-Scale Architecture for Safe, Sustainable Residential Heating Electrification

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Abstract

The global transition away from fossil-fuel heating toward electricity-based systems — primarily heat pumps, supplemented by resistive space heaters — represents one of the largest sustained load-growth events in the history of electrical grids. Unlike other electrification vectors (e.g., vehicle charging, cooking), heating loads exhibit extreme temporal correlation: millions of devices activate simultaneously during cold evenings, creating worst-case peak demand conditions that existing distribution infrastructure was never designed to serve. Despite this, no unified architecture exists for planning, monitoring, and managing heating-specific electrical loads at the city scale. Current demand-response (DR) and smart-grid

standards — including OpenADR 2.0b, IEEE 2030.5, and Matter — treat heating as a generic controllable load, ignoring the unique thermal-comfort constraints, safety floors (freeze protection), and building-physics interactions that govern how heating devices can be curtailed without harming occupants.

This whitepaper introduces the **Urban Heating Backbone Line (U-HBL)**, a four-layer open architecture comprising: (1) a Grid Layer with a Heating Demand Factor (HDF) planning methodology; (2) a Building Layer centered on the **Heating Distribution Bus (HDB)**, a dedicated smart sub-panel for heating circuits; (3) a Device Layer built around **H-Port**, an open interoperability interface for heating appliances; and (4) a Control and Communication Layer that implements temperature-first demand response. U-HBL is accompanied by a phased city deployment framework covering load study, zonal rollout, code integration, and manufacturer certification. Preliminary modeling suggests that temperature-first DR — which curtails heating loads based on indoor temperature surplus rather than raw wattage — can reduce coincident peak heating demand by an estimated 15–30% while preserving occupant comfort above defined safety and baseline floors. U-HBL is published under an Apache 2.0 license as an open, royalty-free standard, inviting contributions from utilities, municipalities, device manufacturers, and the standards community.

1 Introduction

1.1 The Heating Electrification Challenge

Global decarbonization commitments — codified in national climate plans, the European Green Deal, and U.S. federal incentive structures — are driving an unprecedented shift from fossil-fuel heating (natural gas, oil, propane) to electric systems. This transition is not optional for jurisdictions pursuing net-zero targets: space and water heating account for roughly 40–60% of residential energy consumption in cold and temperate climates, and remain overwhelmingly served by combustion appliances.

The electrical grid implications of this transition are severe and qualitatively different from other electrification vectors:

- **Resistive space heaters** are the simplest and cheapest electric heating devices, drawing 1–2 kW each. They are purely resistive (power factor ≈ 1.0), providing no load diversity benefit and converting electrical energy to heat at a 1:1 ratio. In cold-climate housing, a single dwelling may operate 3–6 such devices simultaneously during peak heating periods.
- **Heat pumps** offer 2–4× higher efficiency (COP 2.0–4.5 depending on outdoor temperature and system type) and represent the preferred long-term solution. However, they introduce substantial new electrical demand — a typical residential air-source heat pump draws 2–5 kW — and require backup heating (often resistive) when outdoor temperatures fall below their efficient operating range.
- **Temporal correlation** is the critical differentiator. Unlike EV charging (which can be time-shifted), heating demand is tightly coupled to weather and occupancy. On the coldest evenings of the year, virtually all heating devices in a distribution feeder service area activate simultaneously, producing coincident peak loads that can exceed historical planning assumptions by 50–200%.

Current distribution infrastructure — transformers, feeders, service drops — was sized for historical load profiles dominated by lighting, cooking, and cooling. In many jurisdictions, winter peaks were historically 30–50% below summer peaks. Full heating electrification inverts this relationship, creating winter peaks that exceed summer peaks and stress infrastructure in seasons when utility crews and emergency response are least prepared.

The challenge is not merely one of capacity (which can, in principle, be resolved with capital investment) but of *planning methodology*. Existing load-growth planning treats heating as undifferentiated demand, lacking tools to model heating-specific diversity factors, thermal-mass-driven flexibility, or the comfort constraints that limit how aggressively heating loads can be managed. This methodological gap is the primary motivation for U-HBL.

1.2 The Case for a Heating-Specific Layer

A natural question arises: why not address heating electrification through existing smart-grid and demand-response frameworks? The answer lies in the unique characteristics of heating loads that generic frameworks fail to capture.

Why Generic Capacity Planning Fails

Traditional distribution planning uses aggregate demand forecasts, load-growth percentages, and generic diversity factors. These tools were developed for load mixes dominated by lighting, appliances, and air conditioning — loads with well-understood duty cycles and diversity characteristics. Heating loads break these models in three ways:

1. **Weather-coupled simultaneity:** On design-day cold evenings, heating diversity factors approach 1.0 (near-simultaneous operation), far exceeding the 0.4–0.6 diversity typical of mixed residential loads.
2. **Duration:** Unlike air conditioning (which cycles on and off), heating in poorly insulated buildings may run continuously for 8–14 hours, creating sustained rather than intermittent peaks.
3. **Comfort criticality:** Heating is life-safety infrastructure. Unlike deferrable loads, heating cannot be curtailed below physiological safety thresholds without risking hypothermia, pipe freezing, or health consequences for vulnerable populations.

The Coexistence Model

U-HBL does not assume a single heating technology. Real-world heating electrification will involve a heterogeneous mix of devices: heat pumps for baseline heating, resistive space heaters for supplemental or spot heating, radiant panels, heated floors, and emerging technologies. The architecture must accommodate this mix, providing a common management layer regardless of device type.

Temperature-Based DR as the Missing Paradigm

Existing DR programs operate on a *watt-first* basis: the grid operator requests a specific power reduction (e.g., "reduce consumption by 500 W"), and the device or

aggregator must comply. For heating, this approach is fundamentally flawed because equivalent wattage reductions produce wildly different comfort impacts depending on building insulation, thermal mass, outdoor temperature, and current indoor temperature. A 500 W reduction in a well-insulated apartment with substantial thermal mass may be imperceptible for hours; the same reduction in a poorly insulated home at −15 °C may cause the indoor temperature to fall below safe levels within 30 minutes.

U-HBL introduces *temperature-first DR*: devices declare their current temperature, target temperature, and flexibility band (how far below target they can operate). The grid operator requests temperature setback (e.g., "reduce target by 2 °C for 90 minutes"), and each device independently determines the wattage reduction this implies given its local conditions. This approach inherently respects comfort constraints and distributes curtailment equitably based on each building's physical capacity to absorb it.

Survey of Existing Standards

Standard	Scope	Heating-Specific?	Gap Relative to U-HBL
OpenADR 2.0b	Utility-to-aggregator DR signaling	No	No device-level temperature awareness; no building-layer architecture; no heating diversity planning
IEEE 2030.5	Smart energy profile for end devices	No	Generic load control; no thermal-comfort model; no heating-specific metering
Matter (CSA)	IoT device interoperability	Partial (thermostat cluster)	Consumer-grade; no utility-grid integration; no DR signaling; no planning-layer hooks
Zigbee / Z-Wave	Home automation mesh protocols	No	No grid awareness; no temperature-first DR; proprietary ecosystems
ASHRAE Standard 55	Thermal comfort conditions	Yes (comfort)	Defines comfort but provides no grid-interaction, DR

Standard	Scope	Heating-Specific?	Gap Relative to U-HBL
			signaling, or infrastructure planning framework
EN 15232 / ISO 52120	Building automation impact on energy	Partial	Building-centric; no grid-layer planning; no device interoperability protocol

No existing standard spans the full stack from grid planning through building infrastructure to device interoperability with heating as the organizing principle. U-HBL fills this gap.

1.3 Scope and Objectives

U-HBL is defined as a *parallel planning layer* — it supplements rather than replaces existing electrical codes, grid standards, and building regulations. Its scope and non-goals are defined as follows:

In Scope

- **U-HBL Architecture:** A four-layer reference architecture (Grid, Building, Device, Control) for heating-specific load planning and management.
- **H-Port Sub-Standard:** An open device interface specification enabling heating appliances to declare capabilities and respond to temperature-first DR signals.
- **HDB Sub-Standard:** A specification for a dedicated heating distribution bus (smart sub-panel) within residential buildings, providing isolated metering and control of heating circuits.
- **City Deployment Framework:** A phased methodology for municipalities and utilities to plan, pilot, and scale heating electrification using U-HBL.
- **Open, Royalty-Free Licensing:** All specifications published under Apache 2.0, ensuring no licensing barriers to adoption.

Non-Goals

- **No DIY wiring guidance:** U-HBL does not provide installation instructions for end users. All electrical work must be performed by licensed professionals per applicable codes.
- **No replacement of building codes:** U-HBL references NEC, IEC 60364, and local codes but does not supersede them.
- **No product mandates:** U-HBL defines interoperability interfaces but does not mandate specific products, manufacturers, or technologies.
- **No grid operation:** U-HBL provides planning tools and signaling protocols but does not define how utilities operate their grids.

2 Background and Related Work

2.1 Urban Electrification Trends

The transition from fossil-fuel heating to electric systems is proceeding at different rates across regions, driven by policy mandates, gas infrastructure age, and climate severity. Three case studies illustrate the range of conditions U-HBL must address:

Region / Case Study	Current Heating Mix	Policy Driver	Grid Impact Profile
United Kingdom — Gas boiler phase-out	~85% natural gas, ~5% electric, ~10% other	No new gas boilers after 2035 (proposed); Future Homes Standard	Massive winter peak growth on a grid with limited winter headroom; distribution network reinforcement estimated at £10–20B
Nordic Countries — Heat pump saturation	~40–60% heat pump, ~20–30% district heating, ~10–20% electric resistive	Carbon taxes; existing electric infrastructure; cold climate necessity	High penetration already managed; lessons in grid integration, DR, and thermal storage
U.S. Northeast — Heating mix transition	~40% natural gas, ~25% fuel oil, ~15% electric, ~10%	State-level electrification mandates (NY	Aging gas and oil infrastructure; severe cold-climate

Region / Case Study	Current Heating Mix	Policy Driver	Grid Impact Profile
	propane, ~10% other	CLCPA, MA Clean Energy Standard); federal IRA heat pump incentives	peaks (design temps −15 to −25 °C); significant transformer overload risk

Historical load growth in residential distribution has averaged 1–2% per year in developed economies. Full heating electrification can impose a step-change increase of 30–80% in winter peak demand on individual feeders, depending on housing stock, climate zone, and heating technology mix. This magnitude of change exceeds the planning horizons of most utility capital programs and necessitates a dedicated planning methodology.

2.2 Existing Demand-Response Programs

Utility DR programs have a long history of managing controllable loads, primarily:

- **Water heaters:** Direct load control (DLC) programs have successfully managed electric resistance water heaters for decades, leveraging the thermal storage capacity of hot water tanks to shift load by 1–4 hours.
- **Central HVAC:** Thermostat-based DR (e.g., Nest Rush Hour Rewards, Ecobee eco+) adjusts cooling setpoints during summer peaks, typically by 1–3 °F for 1–3 hours.
- **Pool pumps and irrigation:** Deferrable loads with minimal occupant impact.

These programs share a common limitation when applied to space heating: they assume the load can be interrupted or reduced without immediate comfort impact. Water heaters have significant thermal storage (a 50-gallon tank at 120 °F can defer reheat for hours). Central HVAC in summer benefits from higher thermal mass interaction (warm buildings cool slowly). Space heating in winter, however, operates under more severe constraints — poorly insulated homes can lose 1–2 °C per hour when heating is curtailed, and vulnerable populations (elderly, infants, immunocompromised) face health risks at temperatures below 16–18 °C.

No existing DLC or DR program provides a framework for heating-specific curtailment that accounts for thermal mass, insulation quality, and occupant vulnerability on a per-building basis. U-HBL's temperature-first DR paradigm addresses this gap directly.

2.3 Smart Grid and IoT Standards

The smart grid standards landscape relevant to heating electrification includes:

- **OpenADR 2.0b:** A mature standard for utility-to-aggregator DR signaling. Defines signal types (simple, electricity price, load control) but operates at the aggregator level, not the device level. Does not define device-side behavior, thermal-comfort constraints, or heating-specific parameters.
- **IEEE 2030.5 (Smart Energy Profile 2.0):** Provides end-device communication for energy management, including DER and DR functions. Offers generic load control but lacks heating-specific data models (e.g., indoor temperature, thermal flexibility, priority modes).
- **Matter (Project CHIP / CSA):** The emerging consumer IoT standard includes thermostat and HVAC clusters but is designed for consumer home automation, not utility-grid interaction. No DR signaling, no grid-layer hooks, no planning methodology.
- **ASHRAE Standard 55 (Thermal Environmental Conditions for Human Occupancy):** Defines comfort zones and acceptable temperature ranges. U-HBL references ASHRAE 55 for comfort floor definitions but extends it with operational parameters (safety floor, DR flexibility band, override semantics).

2.4 Heat Pump and Space Heater Market

The global heat pump market is experiencing rapid growth, driven by decarbonization policy and energy cost volatility. Residential heat pump shipments in the EU exceeded 3 million units in 2023, and the U.S. market surpassed 4 million units (including both ducted and ductless systems). The space heater market — comprising portable convection heaters, radiant panels, baseboard heaters, and wall-mounted units — remains large, estimated at 80–100 million units sold globally

per year, owing to low capital cost ($30–300 per unit) and zero installation requirements.

The interoperability gap is stark: virtually no resistive space heaters on the market today support any form of grid communication, demand response, or even basic energy reporting. Heat pumps increasingly support Wi-Fi connectivity but use proprietary apps and APIs with no standardized DR interface. U-HBL's H-Port specification targets this gap, defining a minimal interoperability interface that can be implemented at low cost even in simple resistive devices.

3 U-HBL System Architecture

3.1 Design Principles

The U-HBL architecture is governed by five locked design principles. These principles are immutable within the standard and serve as the evaluation criteria for all architectural decisions:

#	Principle	Statement	Justification
P1	**Heating-Aware Planning**	Grid infrastructure planning shall include heating-specific load parameters (HDF, heating diversity factor, climate margin) as first-class inputs to capacity analysis.	Generic load forecasting fails to capture heating-specific simultaneity, duration, and weather coupling. Without dedicated planning parameters, utilities will systematically under-provision for heating electrification.
P2	**Dedicated Circuits**	Heating loads shall be served by identifiable, independently metered circuits within the building electrical system.	Mixing heating loads with general-purpose circuits makes it impossible to measure, monitor, or control heating demand independently. The HDB provides this isolation.

#	Principle	Statement	Justification
P3	**H-Port Interoperability**	All U-HBL-participating devices shall implement the H-Port interface, declaring their type, rated power, priority, and flexibility to the zone controller.	Without a common device interface, heating management degrades to coarse-grained circuit-level switching, losing the granularity needed for comfort-preserving DR.
P4	**Temperature-First DR**	Demand response signals shall be expressed in temperature terms (setback magnitude and duration), not wattage. Devices determine their own power response based on local conditions.	Watt-first DR ignores building physics and creates inequitable comfort impacts. Temperature-first DR inherently adapts to each building's thermal characteristics.
P5	**Economic Sustainability**	U-HBL participation shall provide measurable economic benefit to both utilities (reduced peak infrastructure cost) and customers (lower bills through flexibility credits).	Standards without economic incentives fail to achieve adoption. U-HBL must demonstrate positive ROI for all participants.

3.2 Grid Layer

The Grid Layer defines how utilities incorporate heating-specific parameters into distribution planning. The central concept is the **Heating Demand Factor (HDF)**, a planning parameter that quantifies the expected peak heating load on a feeder or transformer:

Formula: Heating Demand Factor (HDF)

$$\mathbf{HDF = N_{connected} \times P_{avg_rated} \times DF_{heating} \times CM}$$

Where:

- **$N_{connected}$** = Number of dwellings with electric heating on the feeder
- **P_{avg_rated}** = Average total rated heating power per dwelling (kW)
- **$DF_{heating}$** = Heating diversity factor (0.0–1.0)
- **CM** = Climate margin factor (≥ 1.0)

HDF Parameter Ranges

Parameter	Symbol	Typical Range	Notes
Connected dwellings	$N_{connected}$	1–200 per feeder	From utility GIS and AMI records
Avg rated heating power	P_{avg_rated}	3–15 kW	Depends on dwelling size, insulation, and device mix (heat pump vs resistive)
Heating diversity factor	$DF_{heating}$	0.6–0.95	Approaches 1.0 on design-day conditions; lower in mild weather. Significantly higher than general-load diversity (0.3–0.5)
Climate margin	CM	1.05–1.30	Accounts for extreme cold events beyond design-day, climate change uncertainty, and future load growth
DR credit (optional)	DR_{credit}	0.70–0.90	Reduction factor applied when U-HBL DR is contractually guaranteed; allows reduced infrastructure sizing

Feeder sizing methodology: The HDF is added to the existing non-heating load forecast for each feeder. Transformer and conductor ratings are validated against the combined peak (existing load + HDF). Where the combined peak exceeds rated capacity, the utility triggers an upgrade study. The optional DR_{credit} factor allows

utilities that deploy U-HBL DR to reduce the effective HDF, deferring or avoiding infrastructure upgrades.

3.3 Building Layer

The Building Layer introduces the **Heating Distribution Bus (HDB)** — a dedicated electrical sub-panel (or logical partition of the main panel) that serves exclusively heating loads. The HDB provides three critical functions:

4. **Electrical isolation:** Heating circuits are separated from general-purpose circuits, enabling independent metering and control.
5. **Centralized monitoring:** A single metering point captures total heating consumption and peak demand.
6. **Zone control interface:** The HDB hosts the zone controller, which communicates with H-Port devices and relays DR signals from the utility.

HDB Architecture Diagram

```
┌──────────────────────────────────────────┐          │     UTILITY
METER      │              └──────────────────────────┴──────────────────────────┘
│              ┌──────────────┴──────────────┐              │
MAIN PANEL        │              │  (General-purpose loads)   │
└──────────────┴──────────────┘                      │
Dedicated feeder                        │
┌──────────────┴──────────────┐          │    HEATING
DISTRIBUTION   │              │      BUS (HDB)       │              │
│              │  ┌──────────────────────────┐  │              │ │
Heating Meter / CT  │   │              │  └──────────┴──────────┘   │
│          │            │              │  ┌──────────┴──────────┐   │
│  │  Zone Controller   │   │              │  │  (H-Port gateway)   │   │
│  └──────────┴──────────┘   │              │          │           │
```

```
│   ┌──────────┼──────────┐      │            │   │      │      │      │
│  ┌┴┐   ┌┴┐   ┌┴┐    │            │  │Ckt│  │Ckt│  │Ckt│     │
│  │1│   │2│   │3│    │            │  └┬┘   └┬┘   └┬┘     │
└──┴──────┴──────┴──────┴──────────┘              │      │      │
   ┌┴─────┐  ┌┴─────┐  ┌┴──────┐            │Heat  │  │Space │
   │Radiant│            │Pump   │  │Heater│  │Panel  │            │(H-Port│
   │(H-Port│  │(H-Port│            │device)│  │device)│  │device) │
   └──────┘  └──────┘  └──────┘
```

Retrofit vs New Construction

New construction: The HDB is designed into the electrical plan from the outset, with dedicated conduit runs, appropriately sized conductors, and panel space allocated for heating circuits. Cost increment over standard construction is estimated at $200–500 per dwelling.

Retrofit: An existing home adds a small sub-panel (4–8 circuit spaces) fed from the main panel, with existing heating circuits relocated to the HDB. This is more costly ($700–1,900 per dwelling) but feasible in most residential construction types. The zone controller and metering equipment are installed at the HDB.

Privacy Model

The HDB and zone controller handle potentially sensitive data (heating patterns, occupancy inference, temperature preferences). U-HBL defines a three-tier privacy model:

- **Tier 1 — Aggregate only:** The utility receives only feeder-level aggregate heating load data. No per-dwelling data leaves the building.
- **Tier 2 — Anonymized device:** Per-dwelling data is transmitted but stripped of personally identifiable information. Used for DR and billing.
- **Tier 3 — Full telemetry:** Detailed per-device data is shared with the utility (opt-in only), enabling advanced analytics and research.

3.4 Device Layer

The Device Layer defines the **H-Port** interface — the data contract between a heating device and the zone controller. Every U-HBL-participating device implements H-Port, enabling the zone controller to discover, monitor, and manage heating appliances regardless of manufacturer or technology type.

H-Port Device Declaration Fields

Field	Type	Required	Description
`device_id`	string (UUID)	Yes	Globally unique device identifier, assigned at manufacture
`device_type`	enum	Yes	One of: `heat_pump`, `resistive_convection`, `resistive_radiant`, `baseboard`, `heated_floor`, `towel_rail`, `other`
`rated_power_w`	uint16	Yes	Rated electrical power consumption in watts
`priority`	enum	Yes	One of: `safety`, `baseline`, `luxury`
`flexibility`	object	Yes	Contains `min_temp_c` (floor), `max_setback_c` (max allowable setback), `max_duration_min` (max DR duration)
`status`	object	Yes	Contains `current_power_w`, `current_temp_c`, `target_temp_c`, `mode` (heating / idle / standby / off)
`has_thermostat`	boolean	Yes	Whether the device has an integrated temperature sensor and thermostat
`supports_modulation`	boolean	No	Whether the device can operate at partial power (vs

Field	Type	Required	Description
			on/off only)
`firmware_version`	string	No	H-Port firmware version for compatibility tracking

System → Device Command Set

Command	Parameters	Device Obligation
`set_mode`	`mode`: heating | standby | off	Transition to requested mode. Device MAY refuse `off` if current_temp < safety floor.
`throttle`	`setback_c`: float; `duration_min`: uint16	Reduce target temperature by `setback_c` degrees for `duration_min` minutes. Device MUST NOT reduce below declared `min_temp_c`.
`pre_heat`	`target_c`: float; `by_time`: ISO8601	Raise temperature to `target_c` by specified time. Enables pre-warming before anticipated DR event or peak period.
`report`	`fields`: list of field names	Return current values of requested status fields.

Priority Modes

Priority	Temperature Floor	DR Participation	Use Case
Safety	~4 °C (freeze protection)	Exempt from DR curtailment — device operates as needed to maintain freeze protection	Unoccupied buildings, vacation mode, pipe freeze prevention
Baseline	18–20 °C	Full DR participation with setback limited to `max_setback_c`; floor at 18 °C minimum	Normal occupancy, default operating mode
Luxury	User-defined	Full DR	User-selected

Priority	Temperature Floor	DR Participation	Use Case
	(typically 21–24 °C)	participation; floor remains at user-defined target minus `max_setback_c`	enhanced comfort; first to be curtailed during DR events

3.5 Control and Communication Layer

The Control and Communication Layer connects the Grid, Building, and Device layers through a hierarchical control architecture:

7. **Utility ADMS** (Advanced Distribution Management System) issues heating-specific DR signals based on feeder loading, weather forecasts, and market conditions.
8. **Zone Controller** (at the HDB) receives DR signals, translates them into device-level commands based on local conditions, and orchestrates device responses.
9. **H-Port Devices** execute commands autonomously, respecting their declared flexibility constraints and safety floors.

DR Signaling Flow

1. Utility ADMS detects heating-driven feeder overload risk (forecast or real-time). → 2. ADMS issues zone-level DR signal: `{type: "temperature_setback", setback_c: 2.0, duration_min: 90, priority: "advisory"}`. → 3. Zone controller receives signal, evaluates local conditions (current indoor temperatures, outdoor temperature, time of day). → 4. Zone controller issues per-device `throttle` commands, distributing setback across devices based on priority (luxury devices first, baseline devices last, safety devices exempt). → 5. Devices execute setback, report compliance status. → 6. Zone controller aggregates response and reports to ADMS.

API Surface

Interface	Direction	Protocol	Data Scope
Utility ADMS ↔ Zone Controller	Bidirectional	OpenADR 2.0b extension or REST/MQTT	DR signals, aggregate load reports, compliance status
Building BMS ↔ Zone Controller	Bidirectional	BACnet/IP or REST	Building-level temperature data, occupancy, scheduling
Zone Controller ↔ H-Port Devices	Bidirectional	Local network (Wi-Fi / Ethernet / Thread)	Device declarations, status reports, commands
Public Research API	Read-only	REST (authenticated)	Anonymized, aggregated data for academic and policy research

4 H-Port — Open Device Interface Standard

4.1 Rationale

The decision to define a new device interface rather than extend an existing standard (OpenADR, IEEE 2030.5, Matter) is driven by three factors:

- **Scope mismatch:** OpenADR and IEEE 2030.5 are grid-to-aggregator or grid-to-gateway standards; they do not define device-level behavior for heating appliances. Extending them to include per-device thermal-comfort semantics would require fundamental changes to their data models.
- **Complexity mismatch:** Matter is a comprehensive IoT stack (transport, security, data model, commissioning). Requiring full Matter compliance for a $40 space heater is economically infeasible and architecturally unnecessary when the device communicates only with a local zone controller.
- **Domain specificity:** H-Port is intentionally narrow — it defines only what a heating device must declare and how it responds to temperature-first DR commands. This narrow scope enables low-cost implementation (estimated

BOM increase: $2–8 per device for a basic Wi-Fi or Thread radio plus firmware).

H-Port is designed to be *bridgeable*: a zone controller can translate between H-Port and upstream protocols (OpenADR, IEEE 2030.5, utility-proprietary). This avoids requiring end devices to implement complex grid-facing protocols while preserving interoperability at the building-grid interface.

4.2 Protocol Specification

Message Format Candidates

Format	Pros	Cons	Recommendation
CBOR (RFC 8949)	Compact binary; schema-friendly; IETF standard; low parsing overhead on constrained devices	Less human-readable for debugging; smaller ecosystem than JSON	Primary format for production deployments
JSON (RFC 8259)	Ubiquitous; human-readable; excellent tooling	Verbose; higher parsing cost on constrained MCUs	Required for development/debugging; optional for production
Protobuf (Google)	Extremely compact; fast parsing; strong typing	Proprietary origin; requires code generation; less flexible schema evolution	Allowed but not required; interop testing required

Decision (v0.1): H-Port SHALL support CBOR as the primary wire format and JSON as a mandatory alternate format. Protobuf MAY be supported as an optional third format. The zone controller SHALL accept all supported formats and respond in the format used by the requesting device.

Message Types

- **Declaration (Device → Controller):** Sent on first contact and on any configuration change. Contains all fields from the Device Declaration table (Section 3.4).
- **Status Report (Device → Controller):** Periodic or on-demand report of current operating state (power, temperature, mode).

- **Command (Controller → Device):** One of the four commands defined in Section 3.4 (`set_mode`, `throttle`, `pre_heat`, `report`).
- **Command Ack (Device → Controller):** Acknowledgment with compliance status (`accepted`, `partial`, `refused`) and reason code if refused.
- **Heartbeat (Bidirectional):** Keepalive message at configurable interval (default: 60 seconds). Missed heartbeats trigger timeout handling.

4.3 Device Registration and Declaration

When an H-Port device is connected to an HDB circuit and powered on, it initiates a first-contact protocol:

10. **Discovery:** Device broadcasts an mDNS service advertisement (`_hport._tcp.local`) or sends a multicast announcement on the local network.
11. **Controller handshake:** Zone controller responds with a registration challenge, including controller_id and supported protocol versions.
12. **Capability negotiation:** Device sends its full Declaration message. Controller validates fields, assigns a local slot index, and confirms registration.
13. **Operational:** Device begins periodic status reporting and accepts commands.

Devices that lose connectivity to the zone controller SHALL continue operating in their last-commanded mode. After a configurable timeout (default: 30 minutes), the device SHALL revert to autonomous thermostat control at its declared baseline temperature.

4.4 Demand-Response Signal Handling

Signal Types

Signal Type	Priority Level	Device Obligation	Override Allowed?
Advisory	Low	Device SHOULD comply but MAY	Yes — user can override freely

Signal Type	Priority Level	Device Obligation	Override Allowed?
		ignore. No penalty for non-compliance.	
Mandatory	Medium	Device MUST comply unless doing so would violate its declared safety floor. Non-compliance is logged and reported.	Yes — user override triggers event logging and potential flexibility credit forfeiture
Emergency	Critical	Device MUST comply. Issued only during grid emergency conditions (imminent equipment failure, frequency excursion). Duration limited to 30 minutes per event, 2 events per day maximum.	Safety floor override only — device MUST maintain freeze protection regardless

All signal types preserve the safety floor: no DR signal, regardless of priority, can cause a device to operate below its declared `min_temp_c` (freeze protection threshold).

4.5 Certification and Labeling

Conformance Level	Label	Requirements	Target Devices
H-Port Basic	"U-HBL Declared"	Device implements Declaration and Status Report messages. Responds to `report` command. No DR response required.	Low-cost resistive heaters, simple radiant panels. Enables monitoring and load visibility.
H-Port DR	"U-HBL Ready"	H-Port Basic + responds to `throttle` and `set_mode` commands. Implements advisory and	Mid-range heaters with thermostat, smart baseboard heaters, wall-mounted convectors.

Conformance Level	Label	Requirements	Target Devices
		mandatory DR signal handling.	
H-Port Full	"U-HBL Pro"	H-Port DR + supports `pre_heat`, modulation, emergency DR, and full telemetry reporting. Passes extended conformance test suite.	Heat pumps, smart heating systems, integrated building HVAC.

Test suite concepts (v0.1 outline): Conformance testing will include: (a) message format validation (CBOR/JSON parsing correctness); (b) declaration completeness (all required fields present and valid); (c) command response verification (correct behavior for each command); (d) DR signal compliance testing (advisory, mandatory, emergency scenarios); (e) safety floor enforcement (verify device refuses to operate below `min_temp_c` under all conditions); (f) timeout and fallback behavior (verify safe autonomous operation on controller loss).

5 Heating Distribution Bus (HDB)

5.1 Electrical Design

The HDB is a dedicated electrical sub-panel or logically partitioned section of the main panel that serves exclusively heating loads. Its electrical design parameters are governed by applicable national and regional codes; U-HBL defines the functional requirements:

Parameter	Specification	Notes
Voltage	120/240V split-phase (North America) or 230V single-phase (EU/UK) or 400V three-phase (large buildings)	Matches building service voltage
Bus rating	30–100 A (residential); up to 200 A (multi-family)	Sized per total connected heating load plus 25%

Parameter	Specification	Notes
		margin
Circuit spaces	4–16 (residential); 16–42 (multi-family)	Minimum 2 spare spaces for future expansion
Breaker topology	Individual branch circuit breakers per heating zone or device	15 A or 20 A per circuit (North America); 16 A or 20 A (EU)
Conductor sizing	Per NEC 310.16 / IEC 60364-5-52	Minimum 12 AWG (North America) / 2.5 mm² (EU) for 20 A heating circuits
Feeder from main panel	Sized per NEC 220.82 or IEC equivalent for total heating load	Dedicated feeder breaker in main panel

5.2 Metering and Monitoring

The HDB includes a dedicated metering point to capture heating-specific electrical consumption independently from general building loads. Two implementation approaches are defined:

- **Dedicated meter:** A revenue-grade or sub-meter installed at the HDB feeder, providing independent energy measurement (kWh) and demand measurement (kW). Enables separate heating tariffs and DR settlement.
- **Current transformer (CT) with smart monitoring:** A lower-cost alternative using CTs on the HDB feeder conductors, connected to the zone controller for real-time power monitoring. Suitable for non-revenue applications (load management, analytics) but may not satisfy utility metering requirements.

Data granularity: The metering system SHALL report at minimum 15-minute interval energy data (aligned with utility AMI standards). The zone controller MAY collect higher-resolution data (1-second to 1-minute) for local control and analytics, subject to the privacy tier in effect.

AMI integration: Where the utility deploys Advanced Metering Infrastructure, the HDB meter or CT data SHALL be capable of integration with the AMI headend, enabling the utility to distinguish heating load from total building load in its MDMS (Meter Data Management System).

5.3 Retrofit vs New Construction

Installation Cost Estimation Framework

Component	New Construction	Retrofit	Notes
HDB sub-panel (4–8 space)	$80–200	$150–400	Retrofit requires mounting, feeder run from main panel
Zone controller	$200–500	$200–500	Same hardware; retrofit may require Wi-Fi range extender
Smart outlets / receptacles (per unit)	$40–80	$40–80	H-Port-enabled outlets for plug-in heaters; 2–6 per dwelling typical
Heating meter or CT set	$80–200	$100–300	Revenue-grade meter at high end; CT-only at low end
Wiring and conduit	$100–300	$200–600	Retrofit highly variable based on building construction and routing
Labor (licensed electrician)	Included above	Included above	Assumes 4–8 hours labor for retrofit; 2–4 hours for new construction
Total per dwelling	**$500–1,300**	**$700–1,900**	Excludes heating devices themselves

5.4 Safety and Code Compliance

Protection	NEC Requirement	IEC 60364 Equivalent	U-HBL Additional
Overcurrent	NEC 240 — Branch circuit breakers sized per conductor ampacity	IEC 60364-4-43 — Overcurrent protection coordination	Zone controller monitors per-circuit current; alerts on sustained overload > 80% of breaker rating
Arc Fault (AFCI)	NEC 210.12 —	IEC 62606 (AFDDs)	AFCI/AFDD

Protection	NEC Requirement	IEC 60364 Equivalent	U-HBL Additional
	Required in dwelling unit bedrooms, living areas (jurisdiction-dependent)	— Recommended in some EU jurisdictions	recommended on all HDB branch circuits; required where mandated by local code
Ground Fault (GFCI)	NEC 210.8 — Required in bathrooms, kitchens, laundry, outdoors	IEC 60364-4-41 — RCD protection (30 mA) on socket-outlet circuits	GFCI/RCD on all HDB circuits serving portable plug-in heaters
Over-temperature	Product safety standards (UL, CSA)	IEC 60335-2-30 (room heaters)	H-Port devices SHALL report over-temperature fault to zone controller; controller SHALL disconnect circuit if device reports thermal runaway
Anti-freeze	Not addressed by NEC	Not addressed by IEC 60364	Zone controller SHALL maintain minimum 4 °C on all heating zones regardless of DR state, user settings, or communication loss

Safety Notice

The HDB specification does not replace or supersede any applicable electrical code (NEC, IEC 60364, local amendments). All HDB installations SHALL be designed and installed by licensed electrical professionals in compliance with the authority having jurisdiction (AHJ). U-HBL provides functional requirements that operate within and supplement existing code requirements.

6 Demand Response by Temperature

6.1 Temperature-First vs Watt-First

Dimension	Traditional Watt-First DR	U-HBL Temperature-First DR
Signal unit	Watts or percentage of rated load	Degrees Celsius of setback + duration
Comfort impact	Unpredictable — same watt reduction produces different comfort impacts in different buildings	Bounded — comfort impact directly specified and limited by declared flexibility
Equity	Poor — poorly insulated homes suffer disproportionate comfort loss	Improved — each building's response reflects its thermal capacity
Grid certainty	High — utility knows exact watts curtailed	Moderate — aggregate watt reduction must be estimated from thermal models
Device complexity	Low — device simply modulates power	Moderate — device needs temperature sensor and thermostat logic
Safety risk	Higher — aggressive watt limits can cause unsafe temperatures	Lower — temperature floors are explicit and enforceable
Building physics integration	None — treats building as black box	Implicit — device's thermal response inherently reflects building characteristics
Existing standard support	OpenADR, IEEE 2030.5	U-HBL H-Port (new)

6.2 Thermal Mass and Insulation as Flexibility Assets

A building's capacity to participate in temperature-first DR without comfort violation is determined primarily by its thermal characteristics:

- **Thermal mass:** Buildings with high thermal mass (concrete, masonry) can sustain setback for longer periods because the stored heat in the building fabric releases slowly. A well-insulated concrete apartment building may sustain a 2 °C setback for 3–4 hours with minimal perceived impact.
- **Insulation quality:** The thermal envelope (U-value of walls, windows, roof) determines the rate of heat loss. Better insulation means slower temperature decay during setback, extending the DR window.
- **Air tightness:** Infiltration losses can dominate in older buildings, rapidly dissipating any thermal advantage from setback.

U-HBL treats thermal mass and insulation as *flexibility assets* — building-level characteristics that increase a dwelling's capacity to contribute to DR. Future versions of the standard may define a **Thermal Flexibility Index (TFI)** that quantifies this capacity for each building, enabling the zone controller and utility to optimize DR dispatch across buildings with varying thermal characteristics.

6.3 Minimum Comfort Guarantees

U-HBL defines a strict hierarchy of temperature floors that no DR signal can violate:

Floor	Temperature	Authority	Override Possible?	Rationale
Safety Floor	~4 °C	System-enforced (hardcoded)	No — no signal, command, or user action can disable freeze protection	Prevents pipe freezing, structural damage, and extreme cold hazard in unoccupied spaces
Baseline Floor	18–20 °C	Standard default; adjustable by utility within 16–20 °C range	Utility can lower to 16 °C during emergency events only; user can raise	Aligned with WHO minimum healthy indoor temperature (18 °C); provides adequate comfort for most occupants
User Override	Always available	Occupant	User can always override DR	Preserves occupant

Floor	Temperature	Authority	Override Possible?	Rationale
			setback; override may forfeit flexibility credits	autonomy; critical for trust and adoption

Equity Considerations

Temperature-first DR must not disproportionately burden energy-poor households. U-HBL addresses equity through: (a) the baseline floor guarantee (no occupant can be reduced below 18 °C except in declared emergencies); (b) priority-based dispatch (luxury loads are curtailed before baseline loads); (c) flexibility credit structures that reward participation without penalizing non-participation; (d) exemption mechanisms for medically vulnerable occupants (registered with the utility as critical-needs customers).

6.4 Algorithm Sketch

The following pseudocode illustrates the temperature-first DR decision logic executed by the zone controller for each device during a DR event:

```
FUNCTION handle_dr_event(signal, device):      // Inputs     T_indoor    =
device.status.current_temp_c     T_outdoor   = weather_service.current_temp_c
T_target    = device.status.target_temp_c     T_min       =
device.flexibility.min_temp_c     setback_req = signal.setback_c     duration    =
signal.duration_min     priority    = device.priority      // Step 1: Safety check — never
violate freeze protection     IF T_indoor

<

= 5.0:         RETURN action=EXEMPT, reason="near_safety_floor"      // Step 2:
Priority filtering     IF signal.type == "advisory" AND priority == "safety":
RETURN action=EXEMPT, reason="safety_priority_device"      // Step 3: Calculate
achievable setback     T_new_target = T_target - setback_req     T_floor =
```

```
MAX(T_min, 18.0)   // baseline floor      IF T_new_target

<

T_floor:         T_new_target = T_floor         actual_setback = T_target - T_floor

ELSE:         actual_setback = setback_req      // Step 4: Estimate thermal decay time

decay_rate = estimate_decay_rate(T_indoor, T_outdoor, device)     time_to_floor =

(T_indoor - T_floor) / decay_rate  // hours      IF time_to_floor * 60

<

duration:         // Cannot sustain full duration — reduce duration

adjusted_duration = time_to_floor * 60 * 0.9  // 90% safety margin     ELSE:

adjusted_duration = duration      // Step 5: Issue command     SEND

throttle(         device_id = device.device_id,         setback_c = actual_setback,

duration_min = adjusted_duration     )      RETURN action=THROTTLED,

setback=actual_setback,            duration=adjusted_duration
```

7 Economic Model

7.1 Time-of-Use Pricing Integration

U-HBL enables heating-specific Time-of-Use (TOU) tariffs by providing the metering infrastructure (HDB meter) to distinguish heating consumption from general usage. Proposed TOU structure for heating:

Period	Typical Hours (Winter)	Heating Rate Relative to Flat Rate	Behavioral Incentive
Off-Peak	22:00–06:00	0.5–0.7×	Pre-warm building mass overnight; charge thermal storage

Period	Typical Hours (Winter)	Heating Rate Relative to Flat Rate	Behavioral Incentive
Mid-Peak	06:00–16:00, 20:00–22:00	0.9–1.0×	Normal heating operation during moderate-demand periods
On-Peak	16:00–20:00	1.3–2.0×	Reduce heating during evening peak; rely on pre-warmed thermal mass

The `pre_heat` command in H-Port enables automated off-peak pre-warming: the zone controller instructs devices to raise indoor temperature by 1–3 °C during off-peak hours, building a thermal buffer that is drawn down during on-peak hours when heating is throttled.

7.2 Critical Peak Pricing for Heating

Critical Peak Pricing (CPP) events are triggered when heating-specific grid stress exceeds defined thresholds. Unlike general CPP programs, U-HBL CPP events are:

- **Heating-triggered:** Based on heating load as a share of feeder capacity (e.g., heating load exceeds 70% of transformer rating), not total system demand.
- **Weather-correlated:** Predictable 12–24 hours in advance based on weather forecast, enabling automated pre-warming.
- **Duration-limited:** Maximum 4 hours per event, 10 events per heating season, to maintain customer trust.
- **Temperature-bounded:** CPP setback cannot exceed the device's declared `max_setback_c`; baseline floor remains enforced.

7.3 Flexibility Credits and Bill Discounts

U-HBL defines a flexibility credit system based on **measured flexibility**, quantified in **°C·hours** (the product of temperature setback and duration actually sustained):

Tier	Measured Flexibility (per heating season)	Estimated Bill Discount	Requirements
Bronze	≥ 20 °C·hours	3–5%	H-Port Basic or higher; advisory signal compliance
Silver	≥ 50 °C·hours	5–10%	H-Port DR or higher; mandatory signal compliance ≥ 80%
Gold	≥ 100 °C·hours	10–15%	H-Port Full; mandatory signal compliance ≥ 95%; emergency signal compliance 100%

7.4 Cost-Benefit Analysis Framework for Cities

U-HBL provides a template for city-specific cost-benefit analysis spanning three stakeholder perspectives:

Stakeholder	Cost Categories	Benefit Categories
Utility	IT/OT systems integration, DR program administration, metering infrastructure, customer communications, zone controller subsidies	Deferred transformer and feeder upgrades, reduced peak generation costs, improved load factor, reduced outage risk
Customer	HDB installation cost, H-Port device premium, learning curve	Flexibility credits and bill discounts, improved heating control, participation in sustainability programs
Society	Program administration, equity subsidy funding	Accelerated decarbonization, reduced air quality impacts from fossil heating, grid resilience, job creation (electricians, technology), reduced energy poverty

8 City Deployment Framework

8.1 Phase 1 — Heating Electrification Study

Before deploying U-HBL, a city must understand its current heating landscape and future electrification trajectory. Phase 1 involves:

- **Data collection:** Building stock inventory (age, construction type, insulation level, current heating system), utility load data (feeder-level demand profiles, transformer ratings and loading), weather data (design-day temperatures, heating degree days, historical extremes).
- **Load simulation:** Using the HDF formula (Section 3.2), simulate the impact of partial and full heating electrification on each feeder and transformer. Identify infrastructure bottlenecks and upgrade trigger points.
- **Risk mapping:** Overlay heating electrification load growth on existing infrastructure constraints to identify high-risk zones (transformers that would exceed rated capacity under heating electrification scenarios). Prioritize zones for Phase 2 intervention.

8.2 Phase 2 — Zonal U-HBL Rollout

Prioritization Criteria

Zones are prioritized for U-HBL rollout based on a weighted scoring model:

Criterion	Weight	Rationale
Coldest climate + poorest housing stock	High	Greatest heating load intensity; highest risk of infrastructure overload; most acute comfort and equity concerns
Aging gas/oil infrastructure	High	Areas where fossil heating infrastructure is approaching end-of-life, creating a natural transition point

Criterion	Weight	Rationale
Transformer loading headroom	Medium	Zones with limited headroom benefit most from DR-enabled deferred upgrades
Community readiness	Medium	Active community organizations, existing energy-efficiency programs, willing landlords
Data availability	Low	AMI deployment, building records quality, weather station proximity

Pilot Program Design

- **Scale:** 500–2,000 households per pilot zone, sufficient for statistical significance in DR performance evaluation.
- **Duration:** 1–2 heating seasons (minimum 12 months to capture full seasonal cycle).
- **Metrics:** Peak heating load reduction (%), DR event compliance rate, customer satisfaction (survey), comfort violations (hours below baseline floor), equipment reliability, cost-per-dwelling.
- **Control group:** Comparable zone without U-HBL deployment, with matched metering, to isolate the effect of the intervention.

8.3 Phase 3 — Building Codes and Incentives

Model code language: U-HBL will develop model code language for adoption by jurisdictions, specifying: (a) HDB as a required element in new residential construction with electric heating; (b) minimum circuit allocation for heating in new panel installations; (c) H-Port readiness as a recommended feature in heating appliance procurement specifications.

Incentive structures:

- Utility rebates for HDB installation (targeting 30–50% of retrofit cost).
- Point-of-sale rebates for H-Port-certified heating devices.

- Property tax incentives for buildings meeting U-HBL readiness criteria.
- Low-interest financing for comprehensive heating electrification + HDB retrofit packages.

8.4 Phase 4 — Manufacturer Integration

Following successful pilot evaluation, U-HBL specifications are published for manufacturer adoption:

- **Open publication:** Full H-Port and HDB specifications released under Apache 2.0 license, with reference implementations, test vectors, and compliance tools.
- **Conformance testing:** Establishment of an accredited test facility (or self-certification program for H-Port Basic) where manufacturers can validate their products.
- **Certification labels:** "U-HBL Declared," "U-HBL Ready," and "U-HBL Pro" labels (see Section 4.5) affixed to certified products, enabling consumer and utility recognition.
- **Manufacturer consortium:** Formation of a multi-stakeholder consortium (manufacturers, utilities, municipalities, standards bodies) to govern specification evolution, certification, and branding.

8.5 Equity Considerations

Heating electrification and demand response must not exacerbate energy poverty. U-HBL incorporates the following equity safeguards:

- **Energy poverty screening:** Pilot zones must include demographic analysis. Households spending > 6% of income on energy (or meeting local energy poverty thresholds) are flagged for targeted support.
- **Targeted subsidies:** Full HDB installation cost coverage for income-qualified households. Free H-Port-enabled heaters provided through utility low-income programs.

- **Community engagement:** Multilingual program communications. Community-based organizations engaged as trusted intermediaries. In-person enrollment support for populations with limited digital access.
- **Non-punitive DR:** Low-income participants receive flexibility credits for participation but face no penalties or rate increases for non-participation or DR override.
- **Medical exemptions:** Households with medically vulnerable occupants can register for DR exemption or elevated baseline floors (e.g., 21 °C minimum).

9 Simulation and Modeling Hooks

9.1 Load Simulation Parameters

U-HBL defines a parameterized heating load model for simulation and planning purposes. The model enables utilities and researchers to project heating electrification impacts under various scenarios:

Parameter	Symbol	Mild Scenario	Average Scenario	Design-Day Scenario
Outdoor temperature	T_{out}	0 °C (32 °F)	−10 °C (14 °F)	−20 °C (−4 °F)
Heating diversity factor	$DF_{heating}$	0.60	0.80	0.95
Heat pump COP	COP	3.5	2.5	1.5 (resistive backup)
Resistive heater share	R_{share}	10%	25%	50% (HP backup engaged)
DR participation rate	DR_{part}	N/A (no event)	60%	80%
DR peak reduction	DR_{red}	N/A	15%	25–30%

9.2 Climate Zone Adaptation

U-HBL parameters are adaptable to different climate zones. The following parameters vary by climate and should be calibrated to local conditions:

Parameter	ASHRAE Zone 4 (Mixed)	ASHRAE Zone 5 (Cool)	ASHRAE Zone 6 (Cold)	ASHRAE Zone 7 (Very Cold)
Design outdoor temp	−5 to −10 °C	−10 to −18 °C	−18 to −26 °C	−26 to −38 °C
Heating season length	120–150 days	150–200 days	200–240 days	240–300 days
Climate margin (CM)	1.05–1.10	1.10–1.15	1.15–1.25	1.20–1.30
Thermal flexibility (typical)	High (mild temps, slow decay)	Moderate	Low-Moderate	Low (rapid heat loss)
Recommended pre-heat buffer	1–2 °C	2–3 °C	2–4 °C	3–5 °C

9.3 Scenario Modeling

U-HBL simulation tools should support the following partial-electrification scenarios to model realistic transition pathways:

Scenario	Electrification Rate	Technology Mix	DR Participation	Key Question Addressed
Early Adoption	25%	80% heat pump, 20% resistive	30%	When do first infrastructure constraints appear?
Mid Transition	50%	70% heat pump, 30% resistive	50%	Can DR defer transformer upgrades?
Late Transition	75%	65% heat pump, 35% resistive (incl. backup)	70%	What infrastructure is unavoidable?
Full Electrification	100%	60% heat pump, 40% resistive (incl. backup)	80%	Total grid impact and maximum DR value

10 Discussion

10.1 Limitations and Open Questions

U-HBL v0.1 is a working draft with several unresolved design decisions and known limitations:

- **H-Port encoding finalization:** The choice between CBOR, JSON, and Protobuf as primary wire format is preliminary. Field testing is needed to evaluate parsing performance on low-cost MCUs (< $1 BOM), message size impact on constrained networks (Thread, BLE), and developer ecosystem maturity.
- **Thermal mass estimation:** The temperature-first DR algorithm requires an estimate of each building's thermal decay rate. Determining this accurately without intrusive measurement (blower door tests, energy audits) is an open challenge. Machine learning approaches using historical temperature/power data show promise but require validation.
- **Manufacturer adoption incentives:** The BOM cost increase for H-Port ($2–8 for basic, $5–15 for full) is modest but non-trivial for ultra-low-cost heaters. Market incentives (certification labels, utility procurement preferences) must justify this cost to manufacturers.
- **Multi-family vs single-family:** Multi-family buildings present additional complexity: shared heating infrastructure, split incentives between landlords and tenants, limited tenant control over electrical modifications. HDB retrofitting in multi-family buildings may require building-wide coordination and landlord participation frameworks not yet defined.
- **Grid certainty trade-off:** Temperature-first DR provides less watt-level certainty than watt-first DR. Utilities may require statistical confidence intervals on aggregate watt reduction for planning purposes. Defining these confidence models is ongoing work.

10.2 Relationship to Existing Standards

Standards Body	Relevant Standards	U-HBL Relationship	Potential Submission Path
IEEE	IEEE 2030.5, IEEE 1547, IEEE P2781	U-HBL grid layer complements IEEE 2030.5 DR framework; H-Port extends device-level semantics	IEEE Standards Association (SA) — potential new working group or amendment to IEEE 2030.5
IEC	IEC 61850, IEC 61968/61970 (CIM), IEC TC 57	HDF planning methodology maps to CIM load models; DR signaling aligns with IEC 61968 DR data model	IEC TC 57 WG 21 (interfaces and integration) or new project proposal
NFPA (NEC)	NEC Articles 210, 220, 240, 424	HDB design references NEC for conductor sizing, overcurrent protection, and heating equipment circuits	Code-making panel input for future NEC cycles; informational annex on heating sub-panels
ASHRAE	ASHRAE 55, ASHRAE 90.1, ASHRAE 189.1	Comfort floor definitions reference ASHRAE 55; building energy performance aligns with 90.1	ASHRAE SSPC 55 liaison; potential addendum addressing DR comfort interactions
CSA (Matter)	Matter 1.x thermostat cluster	H-Port is complementary to Matter; zone controller can bridge H-Port devices to Matter ecosystem	Contribution of heating DR cluster definition to Matter specification

10.3 Global Applicability

U-HBL is designed to be adaptable across jurisdictions, but several parameters require localization:

- **Voltage and frequency:** HDB specifications accommodate 120/240V 60 Hz (North America), 230V 50 Hz (EU/UK/AU), and 100V 50/60 Hz (Japan). The architecture is voltage-agnostic above the physical layer.

- **Climate diversity:** The HDF climate margin and thermal flexibility parameters must be calibrated to local climate data. Arctic climates (Zones 7–8) present extreme challenges with very limited DR headroom; Mediterranean climates (Zone 3) may have minimal heating load and limited U-HBL applicability.
- **Regulatory diversity:** Electrical codes, utility regulation, DR program authority, and consumer protection law vary widely. U-HBL provides a reference framework; local deployment requires regulatory mapping and compliance review.
- **Market structure:** Vertically integrated utilities, deregulated markets, and cooperative utilities present different incentive structures for U-HBL adoption. The cost-benefit framework (Section 7.4) accommodates these differences but requires market-specific parameterization.

11 Conclusion and Call to Action

The electrification of residential heating is the largest foreseeable load-growth event facing urban electrical grids. Without a dedicated architecture for planning, monitoring, and managing heating loads, cities face a choice between massive infrastructure over-build (costly and slow) or unmanaged demand growth (risking equipment failure, outages, and inequitable service). Neither outcome is acceptable.

U-HBL offers a third path: a structured, open, interoperable framework that treats heating as a first-class planning concern rather than generic load. Through the Heating Demand Factor (HDF), utilities gain a rigorous methodology for quantifying heating-specific infrastructure needs. Through the Heating Distribution Bus (HDB), buildings gain identifiable, metered, controllable heating infrastructure. Through H-Port, devices gain a common language for declaring capabilities and responding to demand-response signals. Through temperature-first DR, the grid gains a comfort-preserving, equitable mechanism for managing peak heating demand — with an estimated 15–30% peak reduction achievable while maintaining occupant comfort above defined floors.

The path forward requires collaboration across stakeholders. We invite:

- **Cities and utilities** to participate as pilot sites for Phase 1–2 deployment. We are actively soliciting a cold-climate U.S. city and a temperate European city for initial pilot programs targeting the 2027–2028 heating season.
- **Heating device manufacturers** to join a manufacturer consortium for H-Port specification review, reference implementation development, and conformance testing. The target is at least three manufacturers with H-Port Basic products available within 18 months of specification finalization.
- **Standards bodies** (IEEE SA, IEC TC 57, ASHRAE) to evaluate U-HBL for potential adoption as a standards project, amendment, or informational report.
- **Researchers and modelers** to utilize the simulation hooks (Section 9) for independent validation of the temperature-first DR concept and HDF planning methodology.
- **The open-source community** to contribute to reference implementations, test suites, and simulation tools via the project repository.

U-HBL is published under the Apache 2.0 license — open, royalty-free, and available to all. The heating electrification challenge is too large and too urgent for proprietary solutions. We believe that an open standard, collaboratively developed and freely available, is the fastest path to safe, sustainable, equitable heating electrification at city scale.

Contributions, comments, and inquiries: Contact Mirnes Kudić via the project repository or at the coordinates provided in the contribution guide.

Appendices

Appendix A — Glossary of Terms

Term	Definition
U-HBL	Urban Heating Backbone Line. A four-layer open architecture for city-scale planning, monitoring, and management of residential heating electrification.
H-Port	Heating Port. An open device interface standard defining how heating appliances declare capabilities, report status, and respond to demand-response commands from a zone controller.
HDB	Heating Distribution Bus. A dedicated electrical sub-panel (or logical partition) within a building that serves exclusively heating loads, providing isolated metering and control.
HDF	Heating Demand Factor. A planning parameter quantifying expected peak heating electrical demand on a distribution feeder or transformer, incorporating connected loads, diversity, and climate margin.
Heating Diversity Factor	The ratio of actual coincident heating demand to the sum of individual device ratings. Ranges from 0 (no simultaneity) to 1.0 (all devices operating at full rating simultaneously). For heating loads, this factor is significantly higher than general residential load diversity.
Priority Mode	One of three operating priority levels assigned to a heating device: Safety (freeze protection, DR-exempt), Baseline (normal comfort, DR-participating), or Luxury (enhanced comfort, first curtailed).
Flexibility Declaration	An H-Port data structure in which a device declares its willingness and capacity to participate in DR: minimum temperature floor, maximum allowable setback, and maximum DR event duration.
Temperature Floor	The minimum indoor temperature that a heating device or zone controller will maintain under any conditions. The Safety Floor (~4 °C) is system-enforced and non-overridable. The Baseline Floor (18–20 °C) is the minimum comfort target during DR events.
DR Window	The time period during which a demand-response event is active and devices are operating under setback commands. Defined by start time and duration.

Term	Definition
Zone Controller	A hardware/software device installed at the HDB that serves as the gateway between utility DR systems and H-Port devices. It receives DR signals, translates them into device-level commands, and aggregates device responses.

Appendix B — H-Port Protocol Reference (Condensed)

Message Type	Direction	Required Fields	Optional Fields	Response
`declaration`	Device → Controller	device_id, device_type, rated_power_w, priority, flexibility, status, has_thermostat	supports_modulation, firmware_version, manufacturer, model	registration_ack (slot_id, controller_id)
`status_report`	Device → Controller	device_id, current_power_w, current_temp_c, target_temp_c, mode	energy_wh_today, error_code, rssi	None (fire-and-forget) or ack
`set_mode`	Controller → Device	device_id, mode (heating | standby | off)	reason_code	command_ack (accepted | partial | refused, reason)
`throttle`	Controller → Device	device_id, setback_c, duration_min	signal_type (advisory | mandatory | emergency), event_id	command_ack (actual_setback_c, actual_duration_min)
`pre_heat`	Controller → Device	device_id, target_c, by_time (ISO 8601)	reason_code, max_power_w	command_ack (accepted | refused, estimated_ready_time)
`report`	Controller → Device	device_id, fields (list of requested status fields)	None	status_report with requested fields

Message Type	Direction	Required Fields	Optional Fields	Response
`heartbeat`	Bidirectional	sender_id, timestamp	uptime_s, error_count	heartbeat_ack

Appendix C — HDB Wiring Reference (Conceptual)

> **Important Notice**
>
> This appendix provides a **conceptual reference only**. It is NOT a substitute for licensed electrical design. All HDB installations must be designed by a qualified electrical engineer and installed by a licensed electrician in compliance with the National Electrical Code (NEC), IEC 60364, or applicable local electrical code. The authority having jurisdiction (AHJ) must approve all installations.

Conceptual HDB layout for a typical single-family dwelling (North America, 120/240V):

Component	Rating	Description
HDB feeder breaker (in main panel)	40–60 A, 240V	Dedicated two-pole breaker supplying the HDB sub-panel
HDB feeder conductors	8–6 AWG (copper), per NEC 310.16	Four-wire (2 hot + neutral + ground) run to sub-panel location
HDB sub-panel	8–16 space, 60 A bus	NEMA 1 indoor enclosure; wall-mounted near primary heating equipment
Branch circuit 1 — Heat pump	30 A, 240V (per equipment nameplate)	Dedicated circuit for heat pump indoor/outdoor unit
Branch circuit 2 — Bedroom heater(s)	20 A, 120V	Serves 1–2 H-Port-enabled smart outlets for plug-in heaters
Branch circuit 3 — Living area heater(s)	20 A, 120V	Serves 1–3 H-Port-enabled smart outlets for plug-in

Component	Rating	Description
		heaters
Branch circuit 4 — Spare	20 A	Reserved for future expansion (e.g., heated floors, additional zones)
Metering CT set	Sized to feeder	Split-core CTs on feeder conductors, connected to zone controller
Zone controller	5V DC (PoE or USB-C powered)	Mounted inside or adjacent to HDB enclosure; Wi-Fi/Ethernet connected

Appendix D — Example City Rollout Scenarios

Scenario 1: Cold-Climate U.S. City — Utica, New York

Parameter	Value
Population	~62,000 (metro ~290,000)
Climate zone	ASHRAE Zone 6A (Cold, Humid)
Design outdoor temperature	−21 °C (−6 °F)
Heating degree days (HDD65)	~7,100
Current heating mix	~55% natural gas, ~25% fuel oil, ~10% electric, ~10% propane/other
Housing stock	Predominantly pre-1970 wood-frame; moderate insulation; many multi-family 2–4 unit buildings
Utility	National Grid (electricity), National Grid / local gas (gas)
Pilot zone	1,000 dwellings in a mixed-use neighborhood with aging gas infrastructure
HDF estimate (full electrification)	8–12 kW per dwelling × 0.90 diversity × 1.20 climate margin = 8.6–12.9 kW peak per dwelling
Infrastructure risk	High — many distribution transformers at 70–85% loading; full electrification would exceed rated capacity on 40–60% of feeders
DR value estimate	20–30% peak reduction via temperature-first DR could defer $2–5M in transformer

Parameter	Value
	upgrades over 10 years for the pilot zone
Equity note	Median household income ~$35,000; significant energy burden; targeted subsidies essential

Scenario 2: Temperate European City — Milan, Italy

Parameter	Value
Population	~1.4 million (metro ~3.2 million)
Climate zone	ASHRAE Zone 4A equivalent (Mixed-Humid); Italian climate zone E
Design outdoor temperature	−5 °C (23 °F)
Heating degree days (HDD18)	~2,400
Current heating mix	~70% natural gas (centralized boilers), ~15% district heating, ~10% electric, ~5% other
Housing stock	Predominantly masonry/concrete apartment buildings; high thermal mass; moderate insulation (varies by era)
Utility	Enel Distribuzione (electricity), various gas distributors
Pilot zone	2,000 dwellings in a residential district with gas boiler fleet approaching end-of-life
HDF estimate (full electrification)	4–7 kW per dwelling × 0.75 diversity × 1.10 climate margin = 3.3–5.8 kW peak per dwelling
Infrastructure risk	Moderate — Italian residential services typically 3–6 kW contracted; full heating electrification requires service upgrades and potentially 3-phase supply
DR value estimate	15–25% peak reduction; high thermal mass in masonry buildings provides substantial DR headroom (3–5 hour setback tolerance)
Equity note	Centralized building boilers complicate per-dwelling HDB retrofit; building-level HDB with per-apartment zone controllers may be more practical

Appendix E — Economic Model Templates

The following parameter tables provide a starting framework for city-specific utility cost-benefit analysis. Values must be populated with local data.

Template 1: Utility Infrastructure Cost Avoidance

Parameter	Unit	Source	Local Value
Distribution transformer upgrade cost	$ per unit	Utility capital planning	[Enter]
Number of transformers at risk (no DR)	Count	HDF analysis (Phase 1)	[Enter]
Number of transformers deferred (with DR)	Count	HDF analysis with DR credit	[Enter]
Feeder reconductoring cost	$ per mile	Utility capital planning	[Enter]
Miles of feeder at risk	Miles	HDF analysis	[Enter]
DR program annual operating cost	$ per year	Utility DR program budget	[Enter]
Zone controller subsidy cost	$ per dwelling	Utility incentive program	[Enter]
Analysis period	Years	Utility planning horizon	[Enter]
Discount rate	%	Utility WACC	[Enter]
Net present value (avoided cost − program cost)	**$**	**Calculated**	**[Calculate]**

Template 2: Customer Economics

Parameter	Unit	Source	Local Value
HDB installation cost (retrofit)	$ per dwelling	Section 5.3 ranges + local labor rates	[Enter]
Utility rebate / subsidy	$ per dwelling	Program design	[Enter]

Parameter	Unit	Source	Local Value
Net customer cost	$ per dwelling	Installation minus rebate	[Enter]
Annual flexibility credit	$ per year	Section 7.3 tiers	[Enter]
Annual TOU savings (off-peak pre-heat)	$ per year	Rate analysis	[Enter]
Simple payback period	Years	Net cost ÷ annual savings	[Calculate]